

Isaac ROS Image segmentation

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[Overview](#)

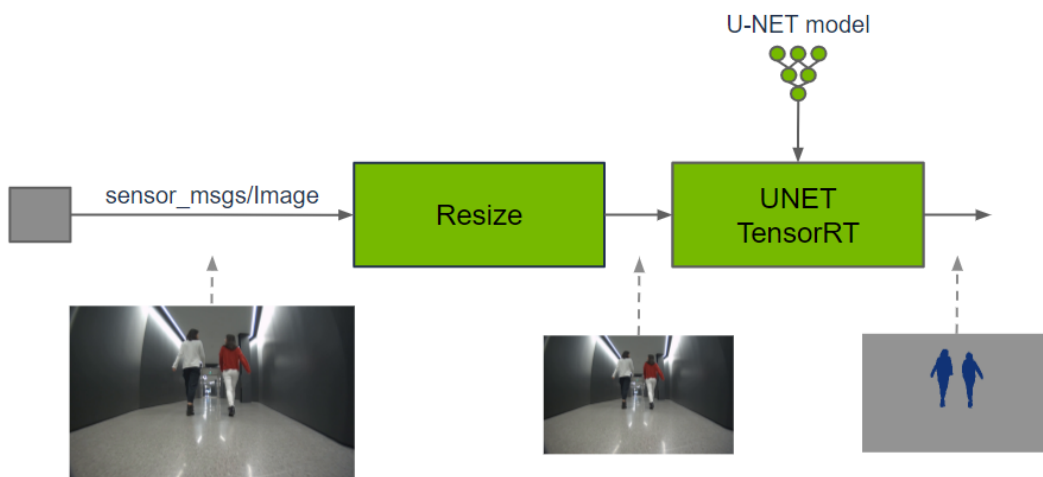
[Quick Experience](#)

Isaac ROS image segmentation official website link: https://nvidia-isaac-ros.github.io/repositories_and_packages/isaac_ros_image_segmentation/index.html

Overview

Isaac ROS Image Segmentation contains ROS packages for semantic image segmentation.

These packages provide methods for classification of an input image at the pixel level by running GPU-accelerated inference on a DNN model. Each pixel of the input image is predicted to belong to a set of defined classes. The output prediction can be used by perception functions to understand where each class is spatially in a 2D image or fuse with a corresponding depth location in a 3D scene.



Quick Experience

To simplify development, we primarily use the Isaac ROS Dev Docker image and demonstrate the effects there. This demonstration does not require any camera device installation; it simulates the data stream from a camera by playing a rosbag file.

Note: If you wish to install on your own device or connect a camera to develop other features, please refer to the Isaac ROS official website and connect to an NVIDIA-specified camera model for your own development.

Open a terminal and enter the working directory.

```
cd ${ISAAC_ROS_WS}/src
```

Enter the Isaac ROS Dev Docker container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following startup command.

```
ros2 launch isaac_ros_examples isaac_ros_examples.launch.py  
••launch_fragments:=segformer  
interface_specs_file:=${ISAAC_ROS_WS}/isaac_ros_assets/isaac_ros_segformer/quick  
start_interface_specs.json model_name:=peoplesensegformer  
model_repository_paths:=[${ISAAC_ROS_WS}/isaac_ros_assets/models]
```

Open a second terminal and enter the container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command:

```
ros2 bag play -l isaac_ros_assets/isaac_ros_segformer/segformer_sample_data
```

View the results

Open a third terminal and enter the container.

```
cd ${ISAAC_ROS_WS}/src/isaac_ros_common && \  
./scripts/run_dev.sh
```

Run the following command to view the results.

```
ros2 run rqt_image_view rqt_image_view /segformer/colored_segmentation_mask
```

